Glossary

Acid: a substance with a pH between 0 (mostly acid) and 7 (neutral).
Adulterated: food that has been prepared or packed in an unsanitary way, contains dirt, could be harmful to health, or may not be correctly labeled.
Aerobic microbes: microbes that grow only with oxygen.
Agar: growth medium for microbes used during testing.
Alkali: a substance with a pH between 7 (neutral) and 14 (mostly alkaline).
Anaerobic microbes: microbes that grow only without oxygen.
Antimicrobial agent: something that destroys or removes microbes.
Aseptic: free from pathogenic microorganisms.
Aseptic packaging/aseptic canning: food and containers are commercially sterilized separately, cooled, and containers filled and sealed in aseptic conditions.
Bacteriostats: substances or conditions that slow growth of bacteria.
Bactericides: substances or conditions that kill bacteria.
Biological control: control of pests or microorganisms using biological methods, e.g., hormones to interfere with reproduction or natural predators.
Biofilm: microcolonies of bacteria that stick to surfaces using a matrix and trap debris and other microbes.
By-product: something other than the main food product that is produced during food processing; may include waste products or wastewater that can be recycled or reused in some way.
Carrier: someone who harbors and releases a pathogen but does not have symptoms of the disease.
CCP: Critical Control Point: a step where control (time, temperature, or protection) can remove a hazard, prevent a hazard, or lessen the risk of a hazard.
Centralized kitchen: one large kitchen facility that prepares foods to be served at several locations.
Chambers test (also called the sanitizer efficiency test): the sanitizer should kill 99.999% of 75 million to 125 million *Escherichia coli* and *Staphylococcus aureus* bacteria within 30 seconds at 20°C (68°F).
Chelating agent (sequestering agent or sequestrant): an inorganic compound that is blended with cleaning compounds to prevent deposits of salts from hard water (i.e., scale).
Chemical poisoning: food poisoning caused by chemicals that are in the food by accident.
CIP: Cleaning in place: the equipment design includes tanks for cleaning and rinse solutions and spray heads and drains inside the equipment to clean it without taking it apart; works well for fluid processing, e.g., dairy and beverages.
Clean: free of visible soil.
Cleaning: removal of soil from a surface.
Colony-forming unit (CFU): units used to count the number of bacteria in a given area or volume; a colony is a collection or group of bacteria derived from one or a few microorganisms.
Commercial sterility: not completely sterile, but enough bacteria and spores are destroyed for the food to be safe.
Communicable disease: infectious disease.

Concentration: the strength of a solution, the amount of compound dissolved in a given volume of liquid.
COP: Cleaning out of place: workers take apart equipment or move it from the processing area to the cleaning area to clean it; works well for small parts and utensils.
Corrosive: eats away, wears away, or weakens a surface.
Critical zone (see danger zone): the range of temperatures at which microbes grow fastest, 2–60°C (36–140°F).
Cross-contamination: carrying soil, microbes, or other particles from a raw-food product to a finished-food product on equipment, utensils, clothing, air, or other vehicles.
Danger zone (see critical zone): the range of temperatures at which microbes grow fastest, 2–60°C (36–140°F).
Delegation: giving someone responsibility for a job.
Disinfection: killing harmful microorganisms.
Dispersion: breaking up material into fine particles that are suspended in solution.
Effluent: waste material discharged from a processing plant.
Enzyme: a biological catalyst that speeds up chemical reactions.
Facultative microbes: microbes that grow either with or without oxygen.
FIFO: first in, first out: principle for rotating stock so that older product is used before the newer inventory.
Food infections: illnesses caused by eating infectious microorganisms, such as bacteria, viruses, or parasites.
Food poisoning: an illness caused by eating food that contains toxins made by microbes or chemical poisons.
Food spoilage: changes in the color, flavor, or texture of foods.
Foodborne infection: illness caused by growth of a foodborne microorganism in a person's body.
Foodborne intoxication: illness caused by toxins produced by microorganisms in food before the food is eaten.
Foodservice: providing food for a large number of people in a commercial operation (restaurant, fast-food operation, or cafeteria) or institution (school, college, military operation, hospital, residential care, prison, etc.).
Fumigant: a vapor or gas used to destroy pests.
Fumigate: using a gas or vapor to destroy pests. A bomb that releases gas is placed in the room, and the room is closed up for several hours to allow the gas to work. The area is well ventilated before people go in again.
Gastroenteritis: inflamed stomach and intestines leading to vomiting, diarrhea, and cramping.
Germicide: chemical that kills microorganisms.
Homogeneous: the same composition or consistency throughout.
Host: the food, animal, or plant that supports growth of a microorganism.
Hygiene: practices used to maintain good health.
Inert gases: gases that do not react with other substances.
Infestation: when pests live somewhere and cause disease and destruction.
Inhibitors: substances or conditions that prevent growth of microbes.
Injunction: a court order that requires a company to correct a problem or stop production.
Inorganic: chemicals or substances that are not made by plants, animals, or other living organisms, e.g., minerals, salts.

Insecticide: chemical poison that kills insects.
Integrated pest management (IPM): effective, environmentally sound, and economical control of pests using a combination of methods (inspection, housekeeping, physical and mechanical methods, biological methods, and chemicals).
Job enrichment: a program to make work more exciting and rewarding for employees.
Low-moisture foods: foods that contain very little moisture, such as bakery goods, nuts and seeds, candy, cereal, grains, and snack foods.
Mechanized: automatic equipment.
Meningitis: inflamed and/or infected membranes around the brain.
Mesophiles: microbes that grow best in moderate temperatures (20-45°C, 68-113°F).
Microbe, microorganism: organisms too small to be seen with the naked eye, such as bacteria, viruses, and some parasites and yeasts.
Microbiology: study of microscopic forms of life, i.e., bacteria, viruses, etc.
Mode of transmission: how the pathogen is carried from the source to the food.
Modified atmosphere packaging (MAP): foods with inert gases inside the sealed package; the gases slow down food spoilage, but pathogens can still grow.
Motivation: giving employees a reason to do their jobs thoroughly.
Mycotoxin: toxin produced by a mold.
Nutrients: substances needed by organisms to grow.
Organic: chemicals or compounds with a carbon skeleton made by plant, animal, or other living beings.
Outbreak: an incident in which two or more people have a similar foodborne illness.
Parasite: an organism that lives off another organism, e.g., parasitic worms that live in the intestines of humans or animals.
Pasteurization: partial sterilization of a food (usually a liquid) at a temperature and for a period of exposure that destroys pathogens without major chemical changes in the food.
Pathogen: a microorganism that can cause disease.
Perishable: a food that spoils or decays quickly or easily, e.g., fruits, vegetables, fresh meat, or milk.
Personal hygiene: maintaining a clean body and clean clothing.
Pest: an animal that causes destruction or disease (e.g., rodents, insects, and birds).
Pesticide: chemical poison that kills pests.
pH: a measurement of acidity and alkalinity, ranging from 0 to 14: values <7 are acidic, 7 is neutral, and values >7 are alkaline.
Pollution/pollutants: contamination of the environment with waste products causing harm to soil, water, plants, and animals.
Porosity: being porous, ability to soak up fluids.
Processing: making a food product from raw ingredients; may include butchering, cleaning, chopping, cooking, mixing, etc.
Productivity: rate of production or amount of food produced per hour.
Psychrotrophs: microbes that grow best in cold temperatures, below 20°C (68°F).
Public health: the protection and improvement of community health by organized community effort, including preventive medicine and sanitary and social science.
Quality assurance: planned and systematic inspection, monitoring, evaluation, and record keeping to maintain an acceptable product.

Quarantine: isolation of a person, animal, or plant to prevent spread of disease.
Quats: quaternary ammonium compounds used as sanitizers.
Ready-to-eat: food that will not be heated or processed again before it is eaten.
Regulations: detailed rules, written after an act has been passed, that specify how and when the law will be carried out.
Relative humidity: availability of water in the atmosphere.
Rodenticide: chemical poison that kills rodents (rats and mice).
Sample: a part of something that represents the whole (e.g., a small amount of milk can be tested to find out about the contents of the whole container).
Sanitary: free of disease-causing microorganisms and other harmful substances.
Sanitation: creating and maintaining hygienic and healthful conditions.
Sanitation audit: formal examination of an operation's sanitation program.
Sanitizer: process or agent (heat, radiation, or chemicals) that kills or removes microbes.
Sanitizing: killing or removing all microbes.
Seafood: edible fish and shellfish.
Shelf life: how long a food product is good to eat before it contains dangerous levels of microbes or the color or flavor goes bad.
Shellfish: water animal with a shell—molluscs (e.g., mussels, clams, snails) or crustaceans (e.g., lobster, shrimp, crab).
Soil: material in the wrong place, often made up of dirt, dust, and specs or scraps of food.
Source: where a pathogen comes from.
***Sous vide* foods:** foods in a strong plastic package from which air is removed (i.e., under vacuum).
Specifications: details of a product or process, e.g., amount, ingredients, temperatures, etc.
Spoilage: decay; off-color, flavor, or odor; or other damage that makes food unacceptable to eat.
Spore: an inactive, resistant, resting, enclosed cell that can produce another growing organism when the conditions are right.
Stability: resistance to chemical, physical, or microbial breakdown.
Statistically valid sample: a large-enough size and number of samples to be sure that it represents the entire batch.
Sterilization: removal or killing of all living organisms, especially microorganisms.
Superheating: heating equipment or food long enough at a high enough temperature to destroy pests, insects, microbes, molds, fungi, and/or spores.
Supervision: making sure employees do each step of their job.
Surfactant: a complex molecule that is blended with a cleaning compound to reduce the energy of the bonds around the soil and allow closer contact between the soil and the cleaning compound.
Suspension: the process of loosening, lifting, and holding soil particles in solution.
Terminal sterilization: the heating step in the traditional canning process. Food is poured into containers, sealed, and then heated to destroy most bacteria.
Thermophiles: microbes that grow best in hot temperatures, above 45°C (113°F).
Total quality management (TQM): a management philosophy that helps managers and employees work together for better productivity, lower costs, and more consistent and safer products.
Toxin: a poisonous chemical.

Ultrapasteurized foods: foods pasteurized at a high temperature for a short time to kill pathogens; the foods are packaged aseptically and may be labeled UHT (ultrahigh temperature).
Vegetative cells: active, living cells.
Wastewater: water contaminated with waste from food processing and cleaning.
Water activity (A_w): availability of water in a food or beverage.
Water hardness: the amount of inorganic salts (such as calcium chloride, magnesium chloride, sulfates, and bicarbonates) in water.
Water softening: removes or inactivates the calcium and magnesium ions in water.
Wetting (penetration): caused by a surfactant that allows the cleaning compound to wet or penetrate the soil deposit and loosen it from the surface.
Wholesome: promoting health or well-being; safe.